

DATA ANALYTICS PROJECT

E-Commerce Sales Analysis

Customer Behavior, Sales Performance & Market Insights



SQL • Power BI • DAX • Data Analysis

Executive Summary

This project analyzes e-commerce sales performance, customer purchasing behavior, sales channels, geographic markets, and revenue trends to identify key business opportunities and areas requiring further investigation.

580

Purchasing
Customers

905

Total Purchases

358.27

Average Order
Value

62.96%

Repeat Revenue
Share

Key Finding

Repeat customers represent 41.21% of purchasing customers but contribute 62.96% of total revenue, highlighting the importance of customer retention and repeat purchasing behavior.

02

Data & Methodology

The analysis follows a structured data analytics workflow, from data preparation and SQL analysis to business intelligence and visualization in Power BI.

01

Data Preparation

Reviewed and prepared the e-commerce sales dataset to ensure the data was suitable for analysis.

02

SQL Analysis

Used SQL queries to calculate sales metrics, customer behavior, channel performance, and geographic performance.

03

Data Modeling

Structured the analytical data model and created calculated measures required for the dashboard.

04

Power BI

Built an interactive dashboard to visualize KPIs, revenue, purchases, customer behavior, and market performance.

Sales Performance Analysis

Sales performance was evaluated across purchasing channels, time, and product categories to identify the main drivers of revenue and purchasing activity.

Revenue by Channel

E-commerce generated higher revenue than App Mobile, indicating a stronger contribution from the e-commerce channel.

Purchases by Channel

Purchase volume was slightly higher through E-commerce than App Mobile, showing a relatively balanced channel mix.

Revenue Trend

Revenue increased from April to May before declining significantly in June, indicating a notable change in monthly sales performance.

Revenue by Product Category

Revenue was distributed almost evenly across the five product categories, with no category showing a clear dominant contribution.

Customer Behavior Analysis

Customer behavior was analyzed through purchasing activity, repeat purchasing patterns, and revenue contribution across different age groups.

Purchasing Customers

The analysis identified 580 purchasing customers, providing the base for evaluating customer activity and repeat behavior.

Repeat Customers

239 customers made repeat purchases, representing 41.21% of purchasing customers.

Repeat Revenue Contribution

Repeat customers generated 62.96% of total revenue, indicating that returning customers contribute disproportionately to sales.

Revenue by Age Range

Customers aged 26–35 generated the highest revenue among the analyzed age groups, while the 56–65 group generated the lowest.

Customer Retention Insight

Repeat customers account for a smaller share of the customer base than one-time purchasers, yet they contribute a much larger share of revenue. This suggests that improving customer retention could have a meaningful impact on revenue performance.

Geographic & Channel Analysis

Geographic and channel performance were compared to identify the strongest markets and understand differences in customer value across purchasing channels.

Revenue by Country

Germany generated the highest revenue among the analyzed markets, followed by France and Italy.

Orders by Country

Germany also recorded the highest number of purchases, followed by France and Italy, indicating strong market activity in these countries.

Average Order Value by Country

Portugal recorded the highest average order value, while Italy recorded the lowest among the analyzed countries.

Channel Performance

E-commerce generated slightly higher revenue and purchase volume than App Mobile, while average order value remained relatively close between the two channels.

Market Insight

Germany is the strongest market by both revenue and purchase volume, while Portugal shows the highest average order value. This suggests that market size and customer value are not necessarily driven by the same factors.

Key Insights & Business Recommendations

The analysis highlights several opportunities to strengthen customer retention, improve channel performance, and focus commercial efforts on high-value customer segments and markets.

Prioritize Customer Retention

Repeat customers represent 41.21% of purchasing customers but contribute 62.96% of total revenue.

Recommendation: Develop retention initiatives such as loyalty programs, personalized offers, and targeted campaigns designed to encourage repeat purchases.

Focus on High-Performing Markets

Germany leads both revenue and purchase volume, making it the strongest market in the current dataset.

Recommendation: Prioritize Germany for customer acquisition and retention efforts while investigating opportunities to improve performance in lower-volume markets.

Investigate the June Revenue Decline

Revenue increased from April to May before declining significantly in June.

Recommendation: Investigate potential drivers such as lower customer activity, seasonality, campaign changes, or incomplete reporting before making strategic decisions.

Target High-Value Customer Segments

Customers aged 26–35 generated the highest revenue, while Portugal recorded the highest average order value.

Recommendation: Test targeted campaigns for high-performing customer segments and evaluate whether similar purchasing behavior can be replicated across other markets.

Business Takeaway

The strongest growth opportunity appears to come from increasing repeat purchasing behavior while maintaining performance in key markets and investigating the sharp decline in June revenue. Retention-focused strategies should therefore be considered alongside targeted market and customer segmentation initiatives.

Conclusion & Next Steps

This analysis provides a clear view of e-commerce sales performance, customer behavior, purchasing channels, and geographic markets.

Key Conclusion

Customer retention represents a significant opportunity, as repeat customers contribute a disproportionately large share of total revenue.

Market Opportunity

Germany is the strongest market by revenue and purchase volume, while Portugal shows the highest average order value.

Further Analysis

The sharp decline in June revenue should be investigated further to identify the underlying business or data-related factors.

Future Work

Future analysis could explore customer segmentation, campaign effectiveness, purchasing frequency, and customer lifetime value.

Final Takeaway

The analysis demonstrates how SQL, DAX, and Power BI can be used together to transform e-commerce transaction data into actionable business insights. The findings provide a foundation for retention-focused strategies, market prioritization, and deeper customer analysis.